

Cultural Interaction Design

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Research Question.

In what ways can emerging technologies transform how we experience, share, preserve, and acquire specifically cultural knowledge in a global society?

Subquestions:

- How do emerging technologies shape the presentation of deeply cultural experiences across virtual, augmented, and physical realities, and what unique affordances does each medium offer for authentic cultural expression?
- In what ways can technologies help bridge linguistic and non-verbal communication barriers, while maintaining cultural nuance and context in cross-cultural conversations?
- What ethical considerations should guide the preservation and sharing of endangered cultural practices through interactive technologies, while encouraging their continued life rather than static archiving?
- How can novel technology promote the learning and sharing of cultural materials and make cross-cultural experiences more accessible in daily life?

Research Theme: Cultural Interaction Design.

I am researching how to design different kinds of cultural interactions in today's globalized and digital age.

Imagine how people interact cross-culturally in the future, how technologies can benefit it, and what designers can or should do. How can they design user experiences for global audiences, especially in new platforms such as XR, wearables, vehicles, or other kinds of ubiquitous interfaces? How can designers understand culture and cultural interaction in the context of design? And how can they find a balance between localization/translation and unification for product and service design?

This research examines the topic of Cultural Interaction Design (CID) from the perspective of cultural studies, media studies, and human computer interaction, and integrates ideas from behavioral science and psychology throughout the research.

Chapter 1. Philosophical Context.

Cultural interaction is studied across academic disciplines, including the fields of cultural studies and media studies. This research also aims to push the boundaries of cultural interaction in today's digital age by exploring how technologies benefit it.

This literature review provides an overview of past scholars' theoretical insights into cultural interactions from the perspective of cultural studies, media studies, and human-computer interactions.

1.1. Defining Cultural Interaction.

Culture is an essential part of people's daily lives. The language we speak, the holidays we celebrate, the traditions we follow, the group of subcultures we belong to, etc. In the early days, culture was viewed as "the complex whole which includes knowledge, belief, art, law, morals, custom, and any other capabilities acquired by man as a member of society" (Edward Tylor, 1871). The more modern explanation of culture from Sociology is "the ways of thinking, the ways of acting, and the material objects that together form a people's way of life" (Macionis, 2019).

Interaction is also a crucial part of people's lives, and this term is frequently used in the field of design, such as "Interaction Design" and "User Interaction." People interact with other people, artifacts, or machines.

So, what is cultural interaction? Past scholars thought of this topic differently. Anthropologist Fernando Ortiz coined the term "transculturation" to describe deculturation, the partial loss of one culture, and neoculturation, the creation of new cultural phenomena. Critical Theorist Homi K. Bhabha defined cultural interaction as a space where cultural differences are expressed and produce new cultural forms, without necessarily losing previous cultures. Sociologist Roland Robertson (1995, 1992) developed the concept of "glocalization" (*"glocal" is a combined word of "global" and "local"*), which describes how global processes are *mediated* through local contexts and encourages people to "think globally, act locally." While their insights are different, they all show that cultural interaction involves multiple cultures that interact with each other and create new cultures.

Cultural interaction is demonstrated in cultural presentation/display, communication/translation, preservation, education/teaching/learning, celebration, and creation. Therefore, it is crucial to think of cultural interaction as multidirectional interactions among people or between people and cultural artifacts.

1.2. Cultural Interaction in Cultural Studies Perspective.

Culture is an essential element of humanity and civilization, and cultural studies, as an interdisciplinary field, has influenced fields including Media, HCI, and psychology.

Culture integrates into people's daily lives. Richard Hoggart described it as an analysis of *everyday* culture, particularly working-class culture, and how it interacts with Mass Media and social change (Hoggart, 1957). Raymond Williams argues that culture is "a whole way of life," which includes *everyday* practices, beliefs, and values (Williams, 1977). Chris Barker also argues that this is a study in relation to *everyday* life (Barker, 2016).

Culture carries and expresses meaning. Cultural Theorist Stuart Hall introduced the theory of representation and meaning (Hall, 1973). Hall argues that representation is central to the production of meaning in culture. He also introduced the model of encoding/decoding in media reception (Hall, 1973), explained that media producers encode messages with meaning, and audiences decode messages based on their own contexts. Barker built on top of Hall's theory of representation by continuing the exploration of how culture is a system of representation. Barker argues that media and cultural texts select, shape, and frame reality (what they present) by constructing meaning, not just showing and reflecting it.

Culture has dimensions. Anthropologist Edward T. Hall (1976) introduced the concept of culture as an iceberg, which consists of Surface Culture and Deep Culture. The former explains that surface elements are visible and easily observable, so it is consciously learned. The latter is the majority of culture that is hidden beneath conscious awareness and implicit, so it is unconsciously learned. Food, language, art, and games are surface culture that are visible or explicit. Cultures that are hidden or implicit, such as concepts of beauty, beliefs, attitudes, and values, are categorized as deep culture. This concept will be considered for user interface & experience design for digital cultural interactions.

Culture plays an important role in globalization. It is a factor that shaped today's globalized world, and culture has evolved because of globalization as well. To see culture from a global perspective, the theory of Cultural Hegemony is a helpful tool to analyze how people learn culture. People tend to be exposed to and be aware of cultures from countries or regions that have a more dominant economic or political status. These dominant cultures are defined by the population groups that are the majority in their countries or regions. On the other hand, some cultures may be ignored by many in the first place and left unsupported by modern media and communication tools. Philosopher and Political Theorist Gramsci (1971) developed the concept of Cultural Hegemony, which argues that dominant groups maintain power primarily through generating cultural consent, in other words, modeling the worldview of the dominant groups as "common sense" or natural. Therefore, it is crucial for culture-related research to think globally and from multiple perspectives instead of solely from the eyes of dominant cultures.

Anthropologist Arjun Appadurai bridged cultural studies with media studies and technology by introducing *The Five Dimensions of Global Cultural Flows* (1990), and three of them—ethnoscapes, mediascapes, and technoscapes—are relevant to cultural interaction with new technologies. Ethnoscapes explain the impact on society by humans of moving groups, such as tourists and immigrants. The movement of people and communications between people from all around the globe shapes today's globalized society. Mediascapes view the capabilities of technological infrastructure that create and spread media content, and Technoscapes provide the tech infrastructure for mediascapes to do so. Technological inventions and innovation led to the creation of traditional analog media (e.g., radio, TV...) and interactive digital media (e.g., video games, Internet, etc.), which are the carriers of cultural expression. Technological advancements also enable the movement of people with rapid transportation and the fast spread of information, which enables and accelerates globalization. While today's digital technologies help bridge cultural barriers, they mainly focus on translation or language education for most people. More importantly, there is a lack of scholarly works and thoughts on culture transmitted – how they were made, expressed, and spread – through technology in the field of cultural studies. As a design researcher, I see room for scholarly development in building design principles or frameworks for digital cultural interaction by combining thoughts from cultural studies with media studies and HCI.

1.3. Cultural Interaction in Media Studies Perspective.

Since the invention and popularization of analog media, people have had more ways to document and spread cultures. People “interact” with these media forms by “consuming” them. They can learn about culture from these media that have images and sounds. Philosopher and Media Theorist Marshall McLuhan (1964) has a leading view of media that bridges it with cultural theory. He called the media an “extension of man,” which extends human capacities and changes the way people see the world. (McLuhan, 1964)

There are, however, challenges. First, not everyone can create and spread cultural content unless they have the right tools for production and broadcast. Second, analog media lack ways or conveniences to revisit or select. People have less exposure to a large variety of cultural content due to the limited choices available in their environment. McLuhan distinguished two kinds of media - “hot” media and “cool” media. In his time, “hot” media, such as radio broadcasts or films, needed less engagement from the audiences. People consume content from these media and absorb cultural knowledge. “Cool” media, however, requires more participation from the audiences to make the communication complete (McLuhan, 1964). Media is a carrier of information, including culture, and how it can communicate that information is also critical to analyze. Media theories by Harold Innis highlight how our civilizations are shaped by communication technologies regarding the dimensions of time and space. Time-biased media are durable but difficult to transport due to their form factor, such as stone carvings and architecture. Space-biased media is the opposite, which is what we mostly consume and use, such as books, newspapers, radio and

television broadcasting, telephone & telegraph, and digital media. The challenge of space-raised media is related to the concept of “Lost Media,” which has recently raised attention on the Internet and has become a subcultural phenomenon. For example, video clips from the early age of the Internet, mostly from the early 90s to the 2000s, were hard to find and considered lost. The same concern has been discussed for other kinds of data, which could either carry or represent cultural phenomena or cultural contents. Communication Theorist Carey discussed this further (1989), emphasizing Innis’s view of how communication technologies affect cultural experiences. McLuhan also famously states, “The medium is the message” (McLuhan, 1964), which argues that communication technologies fundamentally reshape human experiences and sociocultural interactions.

It is also critical to view how the media expresses culture. McLuhan’s “The medium is the message” explained that the way information is communicated has as much impact as the message content itself. Stuart Hall’s encoding/decoding model extends McLuhan’s “the medium is the message” theory by examining how meaning is constructed, interpreted, and circulated through media systems. These views highlighted the important role of media in cultural interaction, including cultural communication and education, from the media’s attributes and capability of cultural expression.

In today’s digital world, media has become more interactive, while traditional media has not been completely replaced but instead evolved. As mentioned, McLuhan distinguished between “hot” media and “cool” media, where the former requires less engagement from the audience, and the latter needs more. McLuhan foresaw the future of interactive digital media.

Digital Media Scholar Janet Murray continued the discussion of digital media by identifying essential properties of digital environments (Murray, 1997) – the *procedural* nature of programmed interactive experiences, *participatory* quality for responding to the audience’s active interactions, *spatial* experiences from immersive environments audience can move and explore, and the *encyclopedic* capacity for vast amount of data in addition to its ability to create complex digital worlds with more detail. Murray’s theory shows the extended possibilities and capabilities of digital media compared to traditional media, highlighting the importance of cultural interaction in interactive digital media. Media disseminates culture, and it also shapes culture in this process. As mentioned in the cultural studies session, Appadurai’s “Five Dimensions of Global Cultural Flows” explains how media technologies produce and spread existing cultural content. Matthew Fuller, an author known in the field of media theory and cultural studies, complemented Appadurai’s theory with Media Ecologies, which highlights that media not only spreads existing culture but also creates new cultural forms.

Existing media theories support the creation, expression, and dissemination of cultural content, but there is currently a lack of discussions and studies explicitly between media and cultural interaction. On top of these media theories, there are opportunities to develop

frameworks or principles for cultural interactions with media types, especially specific digital technologies.

1.4. Cultural Interaction in the HCI Perspective.

Marshall McLuhan said, “*We become what we behold. We shape our tools and thereafter our tools shape us.*” Technological tools being designed and developed, which embody HCI conventions and philosophies, shape people’s lifestyles in many aspects, including cultural education and cultural communication.

Multiple definitions of HCI point out its interdisciplinarity. Fields mentioned included but were not limited to computer science, cognitive science, psychology, design (Ritvo & Allison, 2017), and behavior science (Brey & Søraker, 2009). Cultural studies was not usually listed as one of these multidisciplinary fields contributing to HCI studies, but it has been highlighted as a factor for user analysis (2017) and a valuable contribution for users.

(Sources from an HCI paper collection from ScienceDirect,

<https://www.sciencedirect.com/topics/psychology/human-computer-interaction>)

Several HCI studies explicitly related to culture researched how user interfaces handle cultural factors, such as adaptive interfaces that adjust cultural preferences (Reinecke & Bernstein, 2013), or the opposite—how cultural factors influence users’ behavior, such as Microsoft’s voice assistant research (2018-2022) that examines how cultural factors effect users' comfort. Many such studies focus on adapting cultural factors to the design of user interfaces. However, what about designing for more dimensions of cultural interactions?

The concept of Ubiquitous Computing (UbiComp), or Pervasive Computing, introduced by computer scientist Mark Weiser (1991), envisioned a world in which technologies, like culture, would “weave themselves into the fabric of everyday life until they are indistinguishable from it” (Weiser, 1991). We can observe applications of UbiComp in the implementation of HCI experiences designed to support various kinds of cultural interactions, using different kinds of technologies (AI, Spatial Computing, etc.) and research methods (lifelog, quantified self, etc.)

Computer Scientist Paul Dourish and Cultural Anthropologist Genevieve Bell stand on the foundation of Weiser’s insights by examining the relationship between culture and ubicomp. They define ubicomp as “in practice as much as a social and a cultural phenomenon as it is a technological one” (Dourish & Bell, 2011). Therefore, I see ubicomp as an important practice of Cultural Interaction Design.

1.5. Literature Overview.

These three scholarship areas, cultural studies, media studies, and human-computer interaction, have unique perspectives on culture and cultural interaction. Cultural studies highlighted the importance and meaning of culture and cultural interaction in daily lives and today’s globalized world. The creation, expression, and spread of culture are explained

by ideas from media studies, which is made applied and experiential through theories drawn from HCI. However, there is a lack of specific studies of cultural interaction in these fields, especially when new technologies are involved.

Chapter 2. Empirical Context.

Novel Technologies.

The development, application, and popularization of digital technologies have enhanced cultural interaction in multiple ways.

One central application is language, an essential part of cultural interaction. When we talk about “cross-cultural communication,” many people think of communication across multiple languages. The language barrier is one of the most important factors that affect cultural education and cultural communication. For example, new international students who don’t speak the language spoken in the country or region experience challenges in understanding local culture, learning from courses taught in that language, and communicating with people who do not speak their native language. Globalization encourages cross-cultural interaction in the world of education, international business, and tourism, and there is an increasing need for expanding groups of audiences through linguistic accessibility.

Transcribing text alongside broadcast media was built for accessibility, and it also benefits people who do not master a language. Closed Captioning (CC), which helps people who experience language barriers, has become an industry standard. The service has become accessible to facilities like movie theaters and the majority of consumer electronic products designed for media consumption, such as Sony’s Closed Caption glasses for movie theaters (Figure 1), the built-in caption feature on YouTube, and accessibility features in major operating systems. Content creators are expected to include closed captions for the viewers’ experience, and the advancement of AI technologies makes speech-to-text more accurate so that more content can be transcribed, which results in receiving attention from more audiences.



Figure 1. Sony's subtitled glasses for movie theaters.

Translation takes a step further from transcription. For traditional media, translation allows more audiences to appreciate cultures in languages that they do not speak. The advancement of translation software on mobile platforms makes translation easier, and these capabilities are widely adopted for supporting multilingual communication. One

common use case is when people travel to a foreign country but cannot speak the local language. Digital tools, such as the Google Translate app on smartphones, bridge the language barrier during person-to-person cultural communication. Microsoft’s visual translation tool is able to convert written language in real-time visually, and overlay the result on the video feed from mobile device cameras. In recent years, dedicated translation devices powered by AI have been developed, such as the Vasco Translator, which takes the shape of a pair of Bluetooth earbuds, supports 51 languages, offers real-time translation during conversations, and allows as many as 10 people engaged in a multi-language conversation at the same time.

Besides languages, technologies also help people learn about cultural elements. AI technologies, such as image and sound recognition and Large-Language Models (LLMs), have been used to help people understand culturally specific content. Google Lens and Apple’s Visual Intelligence are trained to recognize cultural artifacts, Shazam is a popular tool used to recognize songs in diverse languages, and the Soundhound app further recognizes a song from a user’s humming of a snippet of a song’s melody; and generative AI powered by LLM can provide more context of a cultural topic for people to understand and generate culturally-specific languages and images.

However, designing new form factors for these technologies can be challenging. Humane, a portable and wearable device marketed as an “AI-pin,” offers an interpreter feature for translation, which provides an intuitive interface for in-person conversation. However, this product was discontinued less than two years after its release.

Another form factor, head-mounted displays (HMDs) and eXtended Reality (XR), has been evolving and becoming a more popular media dissemination and interaction platform. XR is an umbrella term that clusters together the related technologies of AR (Augmented Reality), VR (Virtual Reality), and MR (Mixed Reality). Table 1 provides the definition of these terms and shows the differences between them.

Terms	Definitions
AR (Augmented Reality)	Visualization of digital environment overlaid on real physical environment.
VR (Virtual Reality)	Visualization of a digital, fully-immersive environment that separate from the real physical environment.
MR (Mixed Reality)	A mix of the real physical environment with digital elements where physical elements and digital elements can interact

Table 1. Explaining AR, VR, and MR (Laia, 2022).

XR technologies, especially the AR subset, add a layer of communication in physical spaces by providing more information, expressing information in intuitive ways with different

forms of digital media, and even translating information in the space of the user's presence. XR experiences, such as ARLingo, provide an immersive environment for language learning. A frog-like character plays the role of a person who speaks Japanese and has a conversation with the user. The user can practice Japanese by interacting with this character. This example shows how the user learns basic conversation in Japanese (Figure 2).

Besides languages, there are many cultural elements that we can see, listen to, feel, or even smell and taste, which are representations and expressions of culture, such as music, holidays and traditions, and business etiquette. The ARLingo app not only teaches the language of a culture but also provides an alternative VR immersive visualization of the environment that represents this culture, for example, a hotel in Japan (Figure 3).



Figure 2. A screenshot of the ARLingo app runs on Apple Vision Pro.



Figure 3. ARLingo app runs on Apple Vision Pro in VR mode.

Art Works, Exhibitions, Video Games.

Technology and culture often come together in consumptive and participatory experiences with media. In place of visiting museums in person, XR technology also provides similar experiences for people who have limited access to physical museums by providing virtual and immersive exhibitions. One example is the VR Museum app on Meta Quest, developed by Dracan Works, which shows that the VR platform can replicate a physical cultural touring & learning experience.

Video games as a cultural product are also an important part of modern-day cultural exchange and education. Players can learn about cultures as part of interactive play, often supported by immersive environments and narratives. For example, *Black Myth: Wukong* (《黑神话・悟空》), a 2024 action role-playing game, is a modern expression of the classic Chinese novel *Journey to the West* (《西游记》) and was produced by a Chinese video game developer, Game Science. More importantly, video games have been used to support cultural preservation for endangered cultures, such as *Never Alone (Kisima Inŋitchuŋa)*, a game that centers Inuit culture, aesthetics, and mythology produced in collaboration with tribe members (Figure 4).



Figure 4. Video Game *Never Alone* represents Inuit culture.

Adaptation.

While designing a product or service, an important factor for designers of technologically-supported experiences to think about during the adaptation process is the local market. Recent demographic change in Japan is one of the reasons why some Japanese websites' designs are noticeably different than similar websites from English-speaking countries, as many of them look cluttered by Western web design standards. Due to the phenomenon of an aging society, internet users in Japan consist of a higher percentage of older people, who may prefer an information-dense aesthetic related to Japanese analog media traditions and dislike changes in user interfaces, which are harder for them to relearn (Wada, 2018). One example is Yahoo! Japan's website design, which looks similar to the layout that was commonly used in the late '90s – early 2000s but is still online today (Figure 5).

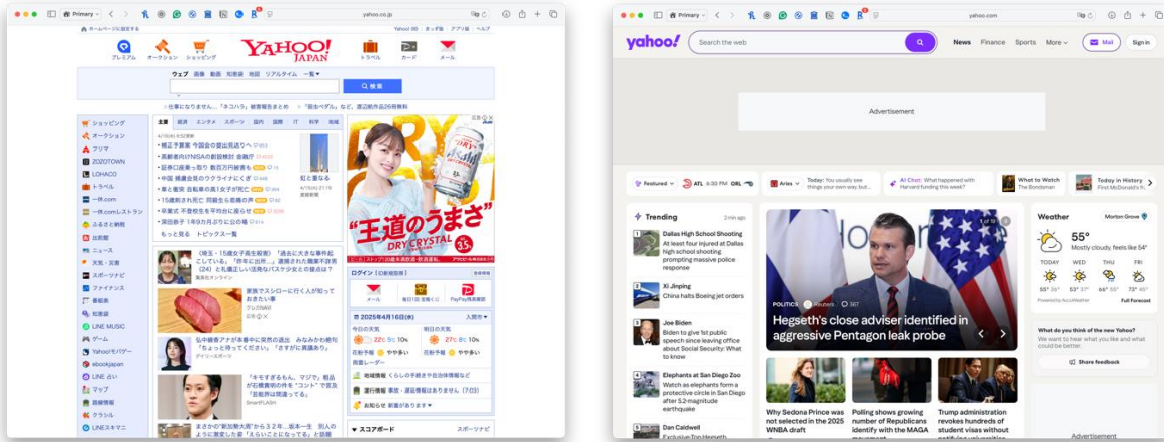


Figure 5. Yahoo! Japan's homepage, as of April 2025, compared to Yahoo! U.S.

Language is also an important factor that affects user experience design. Apple's study of designing user interfaces for the Arabic language is an example of adapting a language that reads from right to left (Figure 6). In this case, Apple suggests that UI elements should be repositioned to follow the right-to-left reading pattern, and they have implemented it in their operating systems.



Figure 6. Apple iOS UI optimized for Arabic.

Adapting products, services, and content for global markets is a common practice by companies, especially large companies that run businesses in international markets. To support the high complexity of serving diverse audiences, several best practices have become standardized recommendations, such as the GILT framework (Table 2). GILT differentiates four main strategies: Globalization (G), Internationalization (I), Localization (L), and Translation (T). This framework is widely used in business.

Term	Definition
Globalization	The process by which businesses or other organizations develop international influence or start operating on a global scale.
Internationalization	The design and development of a product, application or document content that enables easy Localization for target audiences that vary in culture, region, or language.
Localization	The linguistic adaptation of content for the target region, its languages, and cultural particularities in order to make it appropriate, easily understandable, and comfortable to use for local native language speakers.
Translation	The literal word-for-word transformation of a source text into the target language.

Table 2. The GILT Framework (Balaoiu).

Chapter 3. Principles and Methods.

Before discussing what I should do next, I ask myself, how do we value different cultures? Anthropologist and Sociologist Fei Xiaotong (1910-2005) said, 各美其美，美人之美，美美与共，天下大同 (Cherish one's own beauty, respect others' beauty, and when both beauties are respected and cherished, the world will become one.)

Main Method: Prototyping-Based Research.

Since this cultural interaction design research works with novel technologies, the research requires experiments and testing in designing and testing potential solutions in the form of interactive prototypes, such as XR experiences and wearable devices. The products and services mentioned in the last chapter were inspiring, yet there are many opportunities for our research team to explore.

The proposed research program consists of multiple experiments of various kinds of cultural interaction. Each of the experiments will be implemented with interactive prototypes developed for various platforms, including AR, mobile, and wearable devices.

Pilot Study: Qualitative Research Experiment

To gain practice in running studies on the interaction between design and cultural dynamics, in the Fall of 2024, I designed a research experiment entitled “Breaking the Boundaries of Cultural Understanding & Communication Experiences with Digital Technologies” (IRB-2025-44). The experiment studied the ways that people learn and talk about a new culture using different types of digital technologies, and explored the potential best solution for understanding a new culture.

To design this experiment, I conducted qualitative research by running a series of one-to-one interviews to understand how Americans understand an American cultural event, Thanksgiving, and how they spend their time during this holiday (Table 3).

Questions.
Tell me about the way you and your family celebrate /spend time on Thanksgiving.
Besides family traditions, what else do you do to celebrate/spend time on Thanksgiving?
If you are from outside Chicago, do you find the way people celebrate/spend time during Thanksgiving here different from the way you celebrate in your hometown?
How do you feel about celebrating/spending time on Thanksgiving as a kid VS as a grown adult?
Do you celebrate/spend time with friends during Thanksgiving?
Do you celebrate/spend time with other relatives, partners, etc.?
Do you celebrate/spend time on Thanksgiving with people from other countries who are not familiar with American Culture?
(Most Important Question) What do you want people (who are not familiar with American Culture) to learn about Thanksgiving? (Including: Cultural Norm; Personal, family-specific)
What do you suggest people do in order to understand, learn, and experience Thanksgiving? What ways are the best?

Table 3. Interview Questions for my Thanksgiving experiment.

Based on the results from the interviews of 10 participants, I found that people value time with family during the American Thanksgiving holiday, followed by food and traditions. The history of American Thanksgiving, however, was not considered as significant as part of this holiday, and many of the interviewees expressed that they do not favor the history behind this holiday.

I also interviewed 2 international students who have experienced American Thanksgiving in the United States through homestay programs or related events, including “Friendsgiving” gatherings and Thanksgiving parades.

The interview was followed by an experiment to test how international students, especially those with limited knowledge of American Thanksgiving, learned about this holiday with technology support. This experiment took place at a Thanksgiving-related house dinner and party event hosted by ID students. One participant interacted with a launchpad of cultural learning and communication tools in Apple Vision Pro, in order to test how different tools help assist cross-cultural learning and communication in the physical party environment. Two participants also interacted with the XReal Air 2, a pair of AR glasses. One of them is the same person who tested the Vision Pro, who was able to compare the general user experience with Apple Vision Pro in this physical environment. Both participants agreed that the XReal Air 2’s form factor is more acceptable for physical social interaction. Meanwhile, Apple Vision Pro was described as easier to use, according to the participant who tested it. I also reversed roles with these participants by wearing the Vision Pro myself, and I had verbal communication with both of them while discussing American Thanksgiving. While wearing the device, I used the Closed Caption App to transcribe their words, while maintaining eye contact with them, since I could place the live transcription next to their faces. Finally, there are three main criteria for analyzing the data from the experiment results:

- Engagement. From the screen recording, we were able to determine participants’ engagement from the time they spent interacting with the digital devices, and compare it to the time they spent with people and culturally relevant objects .
- Sentiment. We conducted qualitative interviews with selected participants and asked about their impressions of the AR experiment and what they remembered after the experiment.
- Correlation Analysis. We compared the impact of the technologies that were delivered to the participants and interrelated multiple experiential variables.

From this experiment, I learned that conducting an XR experiment was not as smooth as expected since participants can be distracted by other features from the XR platform, plus the surrounding environments. While challenging, the experiment proves that using AR technologies in the form of today’s design in a crowded social environment is not ideal.

For the next steps, I will refine the experiment design by improving the interface and conducting it in different environments, such as learning the Diwali holiday in AR with various media, including music, 3D models, and interactive holiday card design (Figure 7).

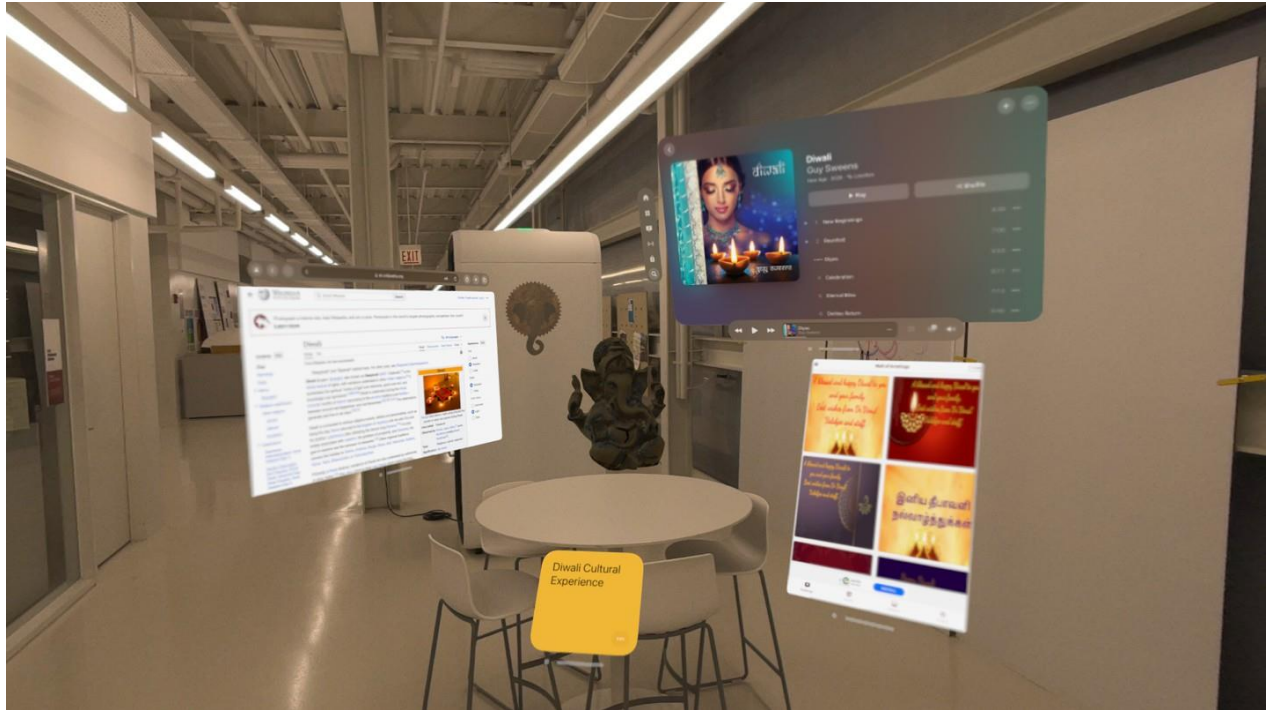


Figure 7. Diwali Cultural experience design prototype on Apple Vision Pro.

In addition to the qualitative holiday culture experiments, I also propose to develop a design tool for creators to generate animated digital humans that represent specific cultures from user input based on multimodal generative AI. My goal is to help creators, especially artists or designers who are not familiar with certain cultures, build animated 3D characters that represent these cultures. By doing this, our approach is to design and develop a tool for creators to generate animated, culturally specific character rigs for digital humans that apply in games, films, TV series, or virtual agents. Using this tool, creators will be able to generate rigs with body language that matches the cultural background of the character. This tool will be powered by our multimodal generative AI trained by culturally specific data collected from videos of various resources. Developing this tool requires powerful computer hardware that is capable of AI training. This tool will enhance productivity and simplify the workflow for creators to complete such tasks. I have submitted the proposal for the Sony Research Award Program in Fall 2024, and it is currently under review.

Research-wise, I plan to continue studying past and current work on various categories of cultural interactions in the philosophical context by conducting Systematic Literature Reviews for each specific cultural interaction. Continuing the literature review will allow me to have a deeper understanding of each type of cultural interaction, which will help me design and develop prototypes and experiments.

Chapter 4. Design Research Framework.

Types of Cultural Interactions.

To understand the contemporary landscape of cultural interactions, we need to provisionally categorize current technological advancements.

Based on what was discussed above, I propose a Cultural Interaction Design Framework (CID Framework) that focuses on a main question: What factors should designers think of when designing different types of cultural interaction using different technologies under different scenarios?

The framework focuses on four specific aspects of cultural interaction: cultural presentation, cultural communication, cultural preservation, and cultural education.

1. Cultural presentation: tell the story of a culture. It is practiced in multiple forms of interactive media, such as video games, interactive arts, and interactive installations in museums.
2. Cultural communications: communications that happen in a group of people or artifacts cross-culturally, which are enhanced with Artificial Intelligence (AI) technologies, such as language translations, voice recognition, and image recognition, plus technologies built with inclusion in mind, such as closed captions as an industrial standard, and localization principles.
3. Cultural education: teach and learn a culture. It has been advanced with the help of technology. For example,
 - a. Interactive language learning platforms, such as Duolingo.
 - b. Immersive language and culture learning apps, such as ARLingo.
4. Cultural preservation: Protecting cultural heritage by teaching it to the next generation and expressing and storing it through technology. For example, video games that express endangered cultures help spread this culture and make it known to more people, especially younger generations.

There are multiple types of Cultural Interaction possible when considering the relationships between people and technological artifacts. Table 4 provides a model for how culture can be made interactive and how designers can more comprehensively participate in the presentation, communication, preservation, and education experiences.

	Person <-> Person	Person <-> Object	Object <-> Object
Cultural Presentation	Demonstration: How can technology assist people to express their	Entertainment: How technology present culture, and how	Data structure: How culture can be translated into data?

	culture, or a cultural content?	people can interact with it?	
Cultural Communication	Translation: How can technology assist people communicate when they speak different language? And how tech help people understand each other cross-culturally?	Cultural Adaptation: How can technology, especially ubiquitous technology, communicate with people by understanding their culture? In other words, how to make tech platforms adapt into target audiences' cultural background, in the aspect of language, habit, etc.?	API: How to connect with tech platforms and let them communicate and exchange data, in order to create new things about culture?
Cultural Education	Teaching and learning: How can technology build a bridge between people to help them teach culture to make it easy to understand?	Learning through AI: How can technology help people understand a culture from artifacts, and this culture's tradition, ritual, and even trend?	AI: How to train an AI to let it understand culture?
Cultural Preservation	Cultural heritage teaching: How can endangered culture be taught to generations and generations?	Cultural heritage expression: How can culture and its stories be expressed by technology?	Datasets: How to store and protect data that represent culture?

Table 4. The Cultural Interaction Design Framework.

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